CUBESAT BASED DARK VESSEL DETECTION PROTOTYPE

SUMMER INTERNSHIP – REPORT

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BONAFIDE CERTIFICATE

It is certified that this report titled as, "**CUBESAT BASED DARK VESSEL DETECTION PROTOTYPE**" is the bonafide work of **MIRUNALINI S (2023105516) , NAVINA P (2023105530)** and **SHANMUGAVALLI P (2023105062) for 7th Semester** who carried out the project work for **Summer Internship**, in the month of June 2025 under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Certificate page

ABSTRACT

Illegal, unregulated or unauthorized maritime vessels frequently disable their identification and communication systems to avoid detection — a behaviour that gives rise to the term "dark vessel." Monitoring such vessels across large ocean regions using conventional, ground-based communication systems alone is difficult, motivating satellite-assisted surveillance approaches. This project presents a low-cost, CubeSat-inspired prototype that demonstrates the core communication and decision logic of a satellite-based dark-vessel detection payload using entirely off-the-shelf embedded hardware.

The prototype is organized into three independent nodes that mirror a real maritime surveillance architecture: a Vessel Node, an onboard Satellite Controller, and a Ground Station. The Vessel Node (Arduino Uno) periodically transmits a heartbeat (HB) packet over an nRF24L01+ wireless link to simulate a vessel broadcasting its identification signal. The Satellite Controller, built around an STM32 "Blue Pill" (STM32F103C8T6) microcontroller, listens for this heartbeat over a fixed time window; presence of a heartbeat is classified as "Vessel Detected," while its absence is classified as "Dark Vessel Detected." In addition to vessel monitoring, the satellite reads an onboard MPU6050 inertial measurement unit to compute its own pitch, roll and yaw — representing the attitude/orientation telemetry a real satellite would report to ground control. The vessel status and orientation values are packed into a compact telemetry string and relayed over a second nRF24L01+ link to a Ground Station (Arduino Uno), which parses and displays the vessel status and orientation angles on a serial monitor in real time.

The complete system was assembled and tested on breadboards for under ₹3,000. Testing confirmed reliable heartbeat-based vessel detection, correct dark-vessel flagging when the heartbeat signal was interrupted, and consistent orientation telemetry delivery to the ground station across both operating conditions. The prototype validates, end-to-end, the sense–process–communicate chain of a satellite-based maritime surveillance system and serves as an accessible educational platform for CubeSat communication concepts, wireless protocol design, and embedded sensor fusion.

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CHAPTER 1

INTRODUCTION

1.1 Project Motivation

Maritime security faces a persistent and growing challenge from unauthorized or "dark" vessels — ships that deliberately disable their Automatic Identification System (AIS) transponders or other identification signals to evade detection during illegal fishing, smuggling, or unregulated activity. Because oceans are vast and largely outside the range of terrestrial radar and communication infrastructure, continuous monitoring of remote maritime regions is difficult using conventional, ground-based systems alone. Satellite-based surveillance offers a scalable solution, but building and testing full-scale satellite payloads is costly and slow.

This project addresses that gap by building a compact, low-cost, CubeSat-inspired prototype that reproduces the essential communication and decision-making pipeline of a satellite-based dark-vessel detection payload — a vessel transmitting a periodic signal, a satellite that listens for that signal and infers vessel status, and a ground station that receives and displays the processed telemetry — using readily available microcontroller and RF hardware.

1.2 Project Overview

The prototype, titled the CubeSat Based Dark Vessel Detection Prototype, models a three-node maritime surveillance chain: a Vessel Node, a Satellite (onboard controller), and a Ground Station. The Vessel Node is implemented on an Arduino Uno and continuously transmits a heartbeat (HB) packet over an nRF24L01+ wireless link, standing in for a vessel's periodic identification broadcast.

The Satellite Controller uses an STM32 "Blue Pill" development board (STM32F103C8T6, ARM Cortex-M3) as its central processor. It listens for the incoming heartbeat over a fixed 5-second window on one nRF24L01+ module; if at least one heartbeat is received within the window the vessel is classified as "Vessel Detected," and if none is received it is classified as "Dark Vessel Detected." The STM32 also interfaces with an MPU6050 six-axis inertial measurement unit over I2C to compute its own pitch, roll and yaw, representing the orientation telemetry a real satellite would downlink alongside its payload data. Vessel status and orientation are packaged into a compact telemetry packet (V, P, R, Y) and transmitted over a second nRF24L01+ link to the Ground Station.

The Ground Station, a second Arduino Uno fitted with its own nRF24L01+ module, receives the telemetry packet, parses the individual fields, and prints the vessel status and orientation angles to a serial monitor in real time, giving an operator a live readout analogous to a mission control display.

1.3 Problem Statement

- Illegal or unidentified vessels often disable their communication and identification systems, making them difficult to track using conventional means.
- Continuous monitoring over large ocean regions is difficult with terrestrial infrastructure alone.
- Reliable, low-latency communication between remote, independently powered nodes is essential for any satellite-assisted monitoring scheme.
- A low-cost, hardware-representative prototype is required to demonstrate the feasibility of satellite-assisted vessel monitoring before committing to a full-scale satellite payload.

1.4 Objectives

- Design a CubeSat-inspired, three-node wireless communication system representing a vessel, a satellite and a ground station.
- Detect vessel presence using periodic heartbeat packets transmitted over nRF24L01+.
- Identify a vessel as "dark" when no heartbeat is received within a defined listening window.
- Measure the satellite's orientation (pitch, roll, yaw) in real time using an MPU6050 IMU.
- Transmit combined vessel-status and orientation telemetry wirelessly to a ground station.
- Display real-time vessel status and orientation data on the ground station's serial monitor.

1.5 Summary

The CubeSat Based Dark Vessel Detection Prototype demonstrates a complete sense–process–communicate chain across three independently powered embedded nodes. A heartbeat-based presence protocol allows the satellite node to distinguish an actively transmitting vessel from a dark one, while onboard IMU sensing adds a realistic orientation-telemetry dimension to the payload. Two independent nRF24L01+ links — vessel-to-satellite and satellite-to-ground — mirror the uplink/downlink structure of an actual satellite communication system. Built from readily available Arduino and STM32 hardware at a total cost of under ₹3,000, the prototype provides a practical, hands-on platform for exploring CubeSat communication concepts, RF protocol design, and embedded sensor fusion.

CHAPTER 2

SYSTEM DESIGN

2.1 System Architecture

The system is organized into three physically separate nodes, each built around a different microcontroller platform and connected wirelessly through nRF24L01+ transceiver modules. Figure 2.1 shows the complete block diagram of the architecture, including the data flow, addressing scheme and the fields carried by each wireless link.

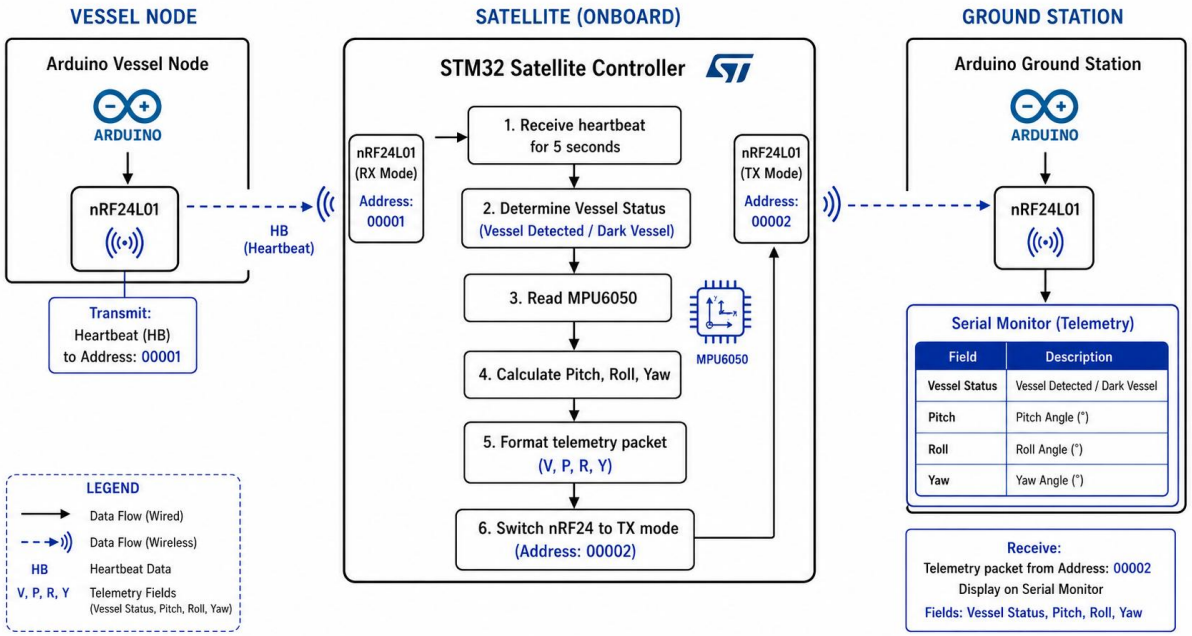


Fig. 2.1 Block Diagram of the CubeSat Dark Vessel Detection Architecture

At the left, the Arduino Vessel Node drives an nRF24L01+ module to transmit heartbeat (HB) packets to address 00001. At the centre, the STM32 Satellite Controller runs the core decision logic: it receives the heartbeat for a 5-second window, determines vessel status, reads the MPU6050, calculates pitch/roll/yaw, formats a telemetry packet, and switches its second nRF24L01+ module into transmit mode to send that packet to address 00002. At the right, the Arduino Ground Station receives the telemetry packet and displays the vessel status and orientation fields on its serial monitor.

2.2 Hardware Components

The prototype was built entirely from low-cost, off-the-shelf components. Table 2.1 lists every component used, along with quantities and approximate cost; the complete three-node system was assembled for under ₹3,000.

Component	Quantity	Unit Price (₹)	Total (₹)
ST-Link Programmer + STM32 Blue Pill (STM32F103C8T6)	1	310	310
Arduino Uno + USB Cable	2	400	800
MPU6050 IMU (Accelerometer + Gyroscope)	1	150	150
nRF24L01+ Transceiver Module (with antenna)	3	175	525
AMS1117 Voltage Regulator	1	100	100
Capacitor (10 µF)	2	2	4
9V Battery	2	25	50
Breadboard	3	33	100
Jumper Wires	50	2	100
Connecting Wires	1 roll	80	80
Power Bank	1	750	750
TOTAL ESTIMATED COST			**₹ 2,969**

Table 2.1 Bill of Materials for the CubeSat Dark Vessel Detection Prototype

STM32 "Blue Pill" (STM32F103C8T6)

The STM32 Blue Pill serves as the Satellite Controller — the central processing node of the system. Programmed via an ST-Link debugger, it was chosen for its ARM Cortex-M3 core, multiple hardware timers, and dedicated SPI and I2C peripherals, which allow it to simultaneously manage two nRF24L01+ radios (one in receive mode, one in transmit mode) and continuously poll the MPU6050 IMU without the timing conflicts that a slower 8-bit microcontroller would introduce.

Arduino Uno (×2)

Two Arduino Uno boards implement the Vessel Node and the Ground Station respectively. Their simplicity and wide library support made them well suited to the comparatively lightweight tasks of periodic heartbeat transmission and telemetry reception/display.

MPU6050 IMU

The MPU6050 is a six-axis inertial measurement unit combining a 3-axis accelerometer and a 3-axis gyroscope, interfaced to the STM32 over I2C. It provides the raw motion data from which the satellite's pitch, roll and yaw are derived, standing in for the attitude-determination sensors of a real satellite.

nRF24L01+ Transceiver (×3)

Three nRF24L01+ modules — one per node — provide the 2.4 GHz wireless links: one link carries the vessel heartbeat to the satellite, and a second, independently addressed link carries telemetry from the satellite to the ground station.

Power and Support Components

Each node is powered independently — 9V batteries and a power bank were used across the vessel, satellite and ground-station breadboards — with an AMS1117 regulator and decoupling capacitors used to supply clean, regulated voltage to the sensitive RF and logic components.

2.3 Communication and Addressing Scheme

Each nRF24L01+ link is configured with a distinct pipe address, so that the satellite's receiver only accepts heartbeat packets from the vessel, and the ground station's receiver only accepts telemetry packets from the satellite. Table 2.2 summarizes the addressing scheme shown in Figure 2.1.

Link	Transmitting Node	Receiving Node (Address)
Vessel → Satellite	Arduino Vessel Node (nRF24L01+)	STM32 Satellite Controller — Address 00001
Satellite → Ground Station	STM32 Satellite Controller (nRF24L01+)	Arduino Ground Station — Address 00002

Table 2.2 nRF24L01+ Link Addressing Scheme

This two-hop, independently addressed design mirrors the uplink (vessel-to-satellite) and downlink (satellite-to-ground) structure of a real satellite communication system, while keeping the prototype simple enough to debug on a breadboard.

CHAPTER 3

SOFTWARE AND CONTROL ARCHITECTURE

3.1 Vessel Node Firmware

The Vessel Node runs a minimal Arduino sketch whose sole purpose is to emulate a vessel's periodic identification broadcast. On startup, it configures its nRF24L01+ radio with a fixed 32-byte payload size, opens a writing pipe to the satellite's address, and stops listening so it operates purely as a transmitter. In the main loop, it formats an incrementing counter into a short text string of the form "HB:<n>" and writes it to the radio once per second, as shown in Figure 3.1.

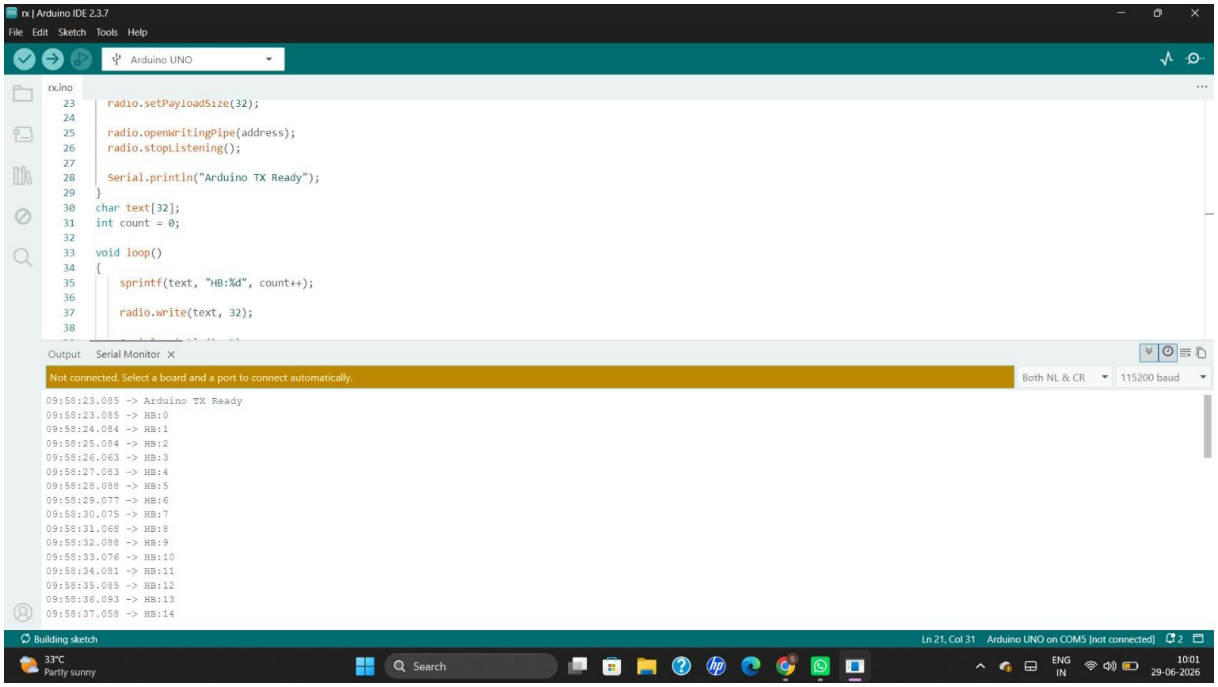


Fig. 3.1 Vessel Node Firmware — Heartbeat Packet Transmission (Arduino IDE)

The Serial Monitor trace in Figure 3.1 confirms the expected behaviour: after printing "Arduino TX Ready," the node emits "HB:0", "HB:1", "HB:2" … at approximately one-second intervals, each timestamped by the IDE at 115200 baud. This steady, predictable cadence is exactly what the satellite's 5-second listening window is designed to detect.

3.2 Satellite Controller Firmware

The STM32 Blue Pill firmware implements the core decision logic of the system, following the six-step sequence shown earlier in the block diagram (Figure 2.1):

- Receive heartbeat packets on the vessel-facing nRF24L01+ radio (RX mode, address 00001) for a fixed 5-second window.

- Determine vessel status: “Vessel Detected” if at least one heartbeat arrived in the window, otherwise “Dark Vessel Detected.”
- Read raw accelerometer and gyroscope values from the MPU6050 over I2C.
- Calculate pitch, roll and yaw from the IMU readings.
- Format a compact telemetry packet carrying vessel status and orientation as V, P, R, Y.
- Switch the second nRF24L01+ radio to TX mode (address 00002) and transmit the telemetry packet to the ground station.

This receive–decide–sense–transmit cycle then repeats continuously, giving the ground station a fresh vessel-status and orientation update roughly every listening cycle.

3.3 MPU6050 Orientation Sensing

On initialization, the STM32 detects the MPU6050 over I2C, wakes it from sleep by writing to the PWR_MGMT_1 register, and performs a calibration pass while the sensor is held stationary to remove accelerometer and gyroscope bias. A representative calibration result recorded during testing was:

Offsets → AccX: 16206, AccY: 3884, AccZ: −17326 | GyroX: −17, GyroY: −214, GyroZ: −55

After calibration, each control cycle reads the raw six-axis values (for example, “Raw AX=−50 AY=8 AZ=16338 | GX=4 GY=5 GZ=0”) and converts them into pitch, roll and yaw angles in degrees, which are then embedded directly in the outgoing telemetry packet alongside the vessel status flag.

3.4 Ground Station Firmware

The Ground Station Arduino configures its nRF24L01+ radio purely as a receiver on the satellite’s telemetry address (00002). Whenever a packet arrives, the firmware parses the comma-separated V, P, R, Y fields and prints a formatted status block to the serial monitor, translating the numeric vessel-status flag into a human-readable label (“VESSEL DETECTED” or “DARK VESSEL”) alongside the pitch, roll and yaw values. Table 3.1 summarizes the fields carried in every telemetry packet.

Field	Description
V (Vessel Status)	1 = Vessel Detected (heartbeat received within the 5-second window); 0 = Dark Vessel Detected (no heartbeat received)
P (Pitch)	Satellite pitch angle in degrees, derived from MPU6050 accelerometer/gyroscope fusion
R (Roll)	Satellite roll angle in degrees
Y (Yaw)	Satellite yaw angle in degrees

Table 3.1 Telemetry Packet Field Definitions (V, P, R, Y)

3.5 Overall Control Flow

Across all three nodes, the complete system follows this end-to-end logic:

- Vessel Node transmits an “HB:<n>” heartbeat packet once per second to the satellite’s address.
- Satellite Controller listens for heartbeats over a 5-second window and classifies vessel status accordingly.
- Satellite Controller reads the MPU6050 and computes pitch, roll and yaw.
- Satellite Controller formats and transmits a V, P, R, Y telemetry packet to the ground station’s address.
- Ground Station receives the packet, parses each field, and displays vessel status and orientation on its serial monitor.
- The cycle repeats continuously, giving a live, near-real-time readout of vessel presence and satellite attitude.

CHAPTER 4

RESULTS AND OBSERVATIONS

4.1 Vessel Heartbeat Transmission

With the Vessel Node powered and running, the Arduino Serial Monitor confirmed "Arduino TX Ready" followed by a steady stream of heartbeat packets ("HB:0", "HB:1", "HB:2" …) arriving at approximately one-second intervals, as shown earlier in Figure 3.1. This validated that the vessel-side transmit chain — payload configuration, writing-pipe setup, and periodic radio.write() calls — functioned correctly and produced a signal representative of a vessel's identification broadcast.

4.2 Satellite Vessel Detection — Normal Condition

With the Vessel Node transmitting normally, the satellite's serial output (Figure 4.1) shows successful MPU6050 detection and calibration, followed by the nRF24L01+ receiver coming online ("NRF RX Ready") and accepting a continuous run of heartbeat packets. After the listening window elapses with heartbeats present, the satellite logs "VESSEL DETECTED," reads the IMU, and prepares its telemetry packet.

```
09:55:17.484 -> Read status = 0
09:55:17.543 -> 3E E4 10 58 FE 2C
09:55:18.023 ->
09:55:18.023 -> ================================
09:55:18.086 -> STM32 MPU6050 Test Starting...
09:55:18.118 -> ================================
09:55:18.151 -> MPU6050 Detected!
09:55:18.183 -> MPU6050 Initialized
09:55:18.183 -> PWR_MGMT_1 = 0x00
09:55:18.730 -> Calibrating MPU6050... DO NOT MOVE THE SENSOR!
09:55:20.184 -> Calibration complete!
09:55:20.216 -> Offsets -> AccX:16206 AccY:3884 AccZ:-17326 | GyroX:-17 GyroY:-214 GyroZ:-55
09:55:20.280 ->
09:55:20.280 -> Starting Clean Readings:
09:55:20.315 ->
09:55:20.315 -> NRF RX Ready
09:55:20.500 -> Received : HB:2
09:55:21.499 -> Received : HB:3
09:55:22.461 -> Received : HB:4
09:55:23.455 -> Received : HB:5
09:55:24.462 -> Received : HB:6
09:55:25.347 -> VESSEL DETECTED
09:55:26.360 -> RAW -50 8 16338
09:55:26.392 -> Raw AX=-50 AY=8 AZ=16338 | GX=4 GY=5 GZ=0
09:55:26.639 -> Sending : V:1,P:0.2,R:0.0,Y:0.0
```

Fig. 4.1 Satellite Serial Log — MPU6050 Initialization and Vessel Detected

Figure 4.2 continues the same trace across several detection cycles, showing the satellite alternating between receiving heartbeats, logging "VESSEL DETECTED," reading fresh raw IMU samples, and sending telemetry such as "Sending : V:1,P:0.1,R:-0.0,Y:-4.6" — confirming that the vessel-detected flag (V:1) was correctly attached to each outgoing orientation reading.

```
09:56:11.510 -> NRF TX Ready
09:56:11.541 -> Telemetry Sent
09:56:11.582 -> NRF RX Ready
09:56:12.579 -> Received : HB:54
09:56:13.565 -> Received : HB:55
09:56:14.573 -> Received : HB:56
09:56:15.547 -> Received : HB:57
09:56:16.573 -> Received : HB:58
09:56:16.573 -> VESSEL DETECTED
09:56:17.618 -> RAW -26 -4 16386
09:56:17.618 -> Raw AX=-26 AY=-4 AZ=16386 | GX=7 GY=23 GZ=-40
09:56:17.902 -> Sending : V:1,P:0.1,R:-0.0,Y:-4.6
09:56:17.937 -> NRF TX Ready
09:56:17.937 -> Telemetry Sent
09:56:17.966 -> NRF RX Ready
09:56:18.546 -> Received : HB:60
09:56:19.555 -> Received : HB:61
09:56:20.584 -> Received : HB:62
09:56:21.559 -> Received : HB:63
09:56:22.551 -> Received : HB:64
09:56:22.997 -> VESSEL DETECTED
09:56:24.017 -> RAW 14 -76 16438
09:56:24.050 -> Raw AX=14 AY=-76 AZ=16438 | GX=-31 GY=5 GZ=19
09:56:24.308 -> Sending : V:1,P:-0.0,R:-0.3,Y:-3.7
```

Fig. 4.2 Satellite Serial Log — Repeated Vessel-Detected Cycles with Telemetry Transmission

4.3 Satellite Dark Vessel Detection

To verify the dark-vessel detection path, the Vessel Node's transmission was interrupted. As expected, once the 5-second listening window elapsed with no heartbeat received, the satellite logged "DARK VESSEL DETECTED" and transmitted telemetry with the vessel-status flag cleared (V:0), while continuing to report valid pitch, roll and yaw values, as shown in Figure 4.3.

```
09:56:37.171 -> NRF TX Ready
09:56:37.171 -> Telemetry Sent
09:56:37.202 -> NRF RX Ready
09:56:42.249 -> DARK VESSEL DETECTED
09:56:43.235 -> RAW 42 36 16370
09:56:43.269 -> Raw AX=42 AY=36 AZ=16370 | GX=26 GY=23 GZ=5
09:56:43.529 -> Sending : V:0,P:-0.1,R:0.1,Y:-1.2
09:56:43.562 -> NRF TX Ready
09:56:43.598 -> Telemetry Sent
09:56:43.646 -> NRF RX Ready
09:56:48.616 -> DARK VESSEL DETECTED
09:56:49.670 -> RAW 30 72 16310
09:56:49.670 -> Raw AX=30 AY=72 AZ=16310 | GX=-11 GY=4 GZ=30
09:56:49.919 -> Sending : V:0,P:-0.1,R:0.3,Y:0.3
09:56:49.984 -> NRF TX Ready
09:56:49.984 -> Telemetry Sent
09:56:50.015 -> NRF RX Ready
09:56:55.024 -> DARK VESSEL DETECTED
09:56:56.062 -> RAW -42 -116 16402
09:56:56.094 -> Raw AX=-42 AY=-116 AZ=16402 | GX=20 GY=-15 GZ=-5
09:56:56.359 -> Sending : V:0,P:0.1,R:-0.4,Y:0.1
09:56:56.391 -> NRF TX Ready
09:56:56.425 -> Telemetry Sent
09:56:56.425 -> NRF RX Ready
09:57:01.443 -> DARK VESSEL DETECTED
```

Fig. 4.3 Satellite Serial Log — Dark Vessel Detected on Loss of Heartbeat

This confirms that vessel-status classification is driven purely by the presence or absence of heartbeat packets within each listening window, independent of the satellite's own orientation-sensing function, which continued to operate normally throughout.

4.4 Ground Station Display

The Ground Station's serial monitor displayed each incoming telemetry packet as a formatted status block. Under normal conditions, with the vessel transmitting, the display read as shown in Figure 4.4.



Fig. 4.4 Ground Station Display — Normal Condition (Vessel Detected)

When the vessel's heartbeat was withheld, the ground station correctly displayed the dark-vessel status, along with the satellite's current orientation readings, as shown in Figure 4.5.



Fig. 4.5 Ground Station Display — Dark Vessel Detected

Across repeated tests, the ground station consistently reflected the satellite's classification within one telemetry cycle, confirming reliable end-to-end delivery of both the vessel-status flag and the orientation data over the second nRF24L01+ link.

4.5 Final Hardware Assembly

Figure 4.6 shows the fully assembled prototype, with the three nodes — Ground Station, Satellite and Vessel — laid out on breadboards, each with its own nRF24L01+ module, power supply and (for the satellite) MPU6050 sensor.

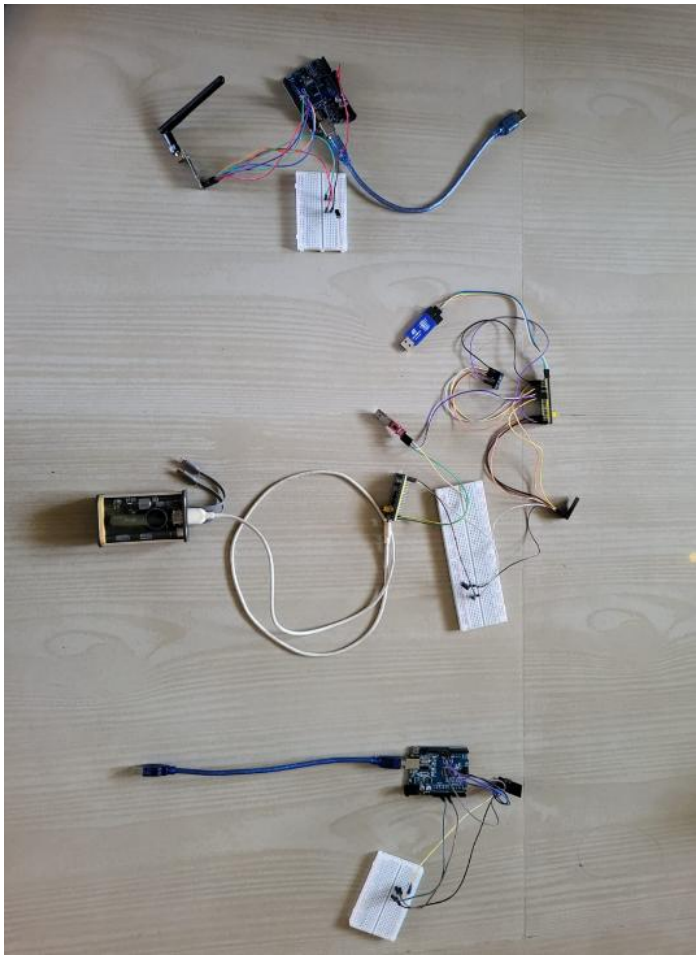


Fig. 4.6 Final Assembled Circuits — Ground Station, Satellite and Vessel Nodes

4.6 Limitations and Future Improvements

While the prototype successfully validated the core heartbeat-detection and telemetry pipeline, several limitations remain to be addressed in future iterations:

- The 5-second listening window is fixed in firmware; an adaptive or configurable window would better balance detection latency against false negatives caused by transient signal loss.
- The current setup uses direct point-to-point nRF24L01+ links rather than a networked or mesh protocol, limiting scalability to a single vessel and a single satellite node.
- Orientation data is derived from a single MPU6050 without GPS or magnetometer fusion, so absolute heading and position are not yet available.
- The system has only been tested at short range on a benchtop; real deployment would require RF range/link-budget testing and a ruggedized enclosure.
- Future versions could add support for multiple simultaneous vessel nodes with unique IDs, GPS-based vessel positioning, and a graphical ground-station dashboard in place of the raw serial monitor output.

CHAPTER 5

CONCLUSION

The CubeSat Based Dark Vessel Detection Prototype demonstrates that the essential communication and decision-making pipeline of a satellite-based maritime surveillance payload can be reproduced, end-to-end, using low-cost and readily available embedded hardware. By dividing the system into three independent nodes — a vessel transmitting a periodic heartbeat, a satellite controller that listens for that heartbeat while sensing its own orientation, and a ground station that receives and displays the resulting telemetry — the project mirrors the uplink/downlink structure and sense-and-report responsibilities of a real satellite payload at a fraction of the cost and complexity.

Testing confirmed that the system reliably distinguishes an actively transmitting vessel from a dark one based purely on heartbeat presence within a fixed listening window, that satellite orientation (pitch, roll, yaw) is computed and transmitted correctly regardless of vessel status, and that the ground station consistently and accurately displays both pieces of information in real time. The STM32 Blue Pill proved capable of concurrently managing two independently addressed nRF24L01+ radios and continuous MPU6050 polling, validating it as a suitable low-cost stand-in for a satellite's onboard processor in an educational or early-prototype context.

Learning Outcomes

- Wireless Protocol Design: Configured and addressed multiple nRF24L01+ radio links for independent uplink and downlink communication paths.
- Embedded Sensor Interfacing: Interfaced and calibrated an MPU6050 IMU over I2C on an STM32 platform to derive real-time pitch, roll and yaw.
- Multi-Node System Integration: Coordinated three independently powered microcontroller nodes (two Arduino Uno, one STM32 Blue Pill) into a single cooperating system.
- Presence-Detection Logic: Implemented a heartbeat-based, time-windowed detection scheme to classify node status from packet arrival patterns.
- Telemetry Packet Design: Designed a compact, human-readable telemetry format (V, P, R, Y) suitable for both machine parsing and direct serial-monitor readability.

Future Scope

- Multi-Vessel Support: Extend the heartbeat protocol with unique vessel IDs so a single satellite node can monitor several vessels simultaneously.
- GPS Integration: Add GPS modules to the vessel and/or satellite nodes to attach real position data to the telemetry stream.
- Long-Range Testing: Characterize the nRF24L01+ links' effective range and reliability under realistic outdoor and marine RF conditions.
- Ground-Station Dashboard: Replace the raw serial monitor output with a graphical dashboard (e.g., a PC or web application) for operator-friendly visualization.
- Autonomous Alerting: Add automated alerts (buzzer, LED, or notification) at the ground station when a dark vessel is detected, reducing reliance on constant manual monitoring.

Overall, the project bridges foundational embedded-systems and wireless-communication concepts with a real-world maritime-security use case, offering both a working proof of concept and a practical platform for further research into low-cost CubeSat-based monitoring systems.

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